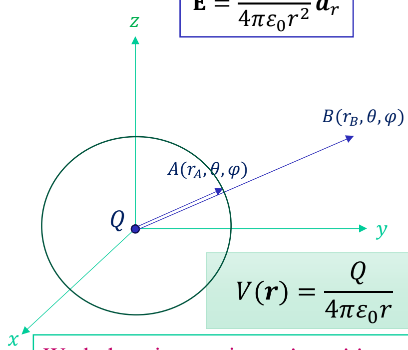
Absolute Potential and Potential Field

Problem:

Find the absolute potential in the presence of field due to point charge Q.

$$\mathbf{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\boldsymbol{a}_r$$



$$V(\boldsymbol{r}) = \frac{Q}{4\pi\varepsilon_0 r}$$

Work done in carrying *unit positive charge* from infinite to field point in the presence of point charge Q.

$$V_{AB} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{r_B}\right)$$

Potential is a Scalar Field.

If r_B recedes to ∞,

$$V_{A\infty} = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r_A} - \frac{1}{\infty}\right)$$

$$V_A = \frac{Q}{4\pi\varepsilon_0 r_A}$$

Equipotential Surface: The locus of all the points having same potential.

V_A is known as Absolute Potential with reference at *infinity*.

$$V_{AB} = V_A - V_B$$

D4.4. **D4.5.**

D4.4. An electric field is expressed in rectangular coordinates by $\mathbf{E} = 6x^2\mathbf{a}_x + 6y\mathbf{a}_y + 4\mathbf{a}_z$ V/m. Find: (*a*) V_{MN} if points M and N are specified by $M(2, 6, -1)$ and $N(-3, -3, 2)$; (*b*) V_M if $V = 0$ at $Q(4, -2, -35)$; (*c*) V_N if $V = 2$ at $P(1, 2, -4)$.

$$E(x, y, z) := \begin{pmatrix} 6x^2 \\ 6y \\ 4 \end{pmatrix}$$

a) Find Vmn

$$M := \begin{pmatrix} 2 \\ 6 \\ -1 \end{pmatrix} \quad N := \begin{pmatrix} -3 \\ -3 \\ 2 \end{pmatrix} \quad V_{MN} := -\int_{N_0}^{M_0} 6x^2\,dx - \int_{N_1}^{M_1} 6y\,dy - \int_{N_2}^{M_2} 4\,dz \quad V_{MN} = -139$$

b) Find Vm if V=0 at Q(4,-2,-35)

$$Q := \begin{pmatrix} 4 \\ -2 \\ -35 \end{pmatrix} \quad V_M := -\int_{Q_0}^{M_0} 6x^2\,dx - \int_{Q_1}^{M_1} 6y\,dy - \int_{Q_2}^{M_2} 4\,dz \quad V_M = -120$$

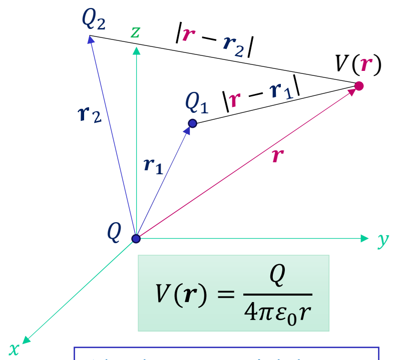
c) Find Vn if V=2 at P(1, 2, -4)

$$P := \begin{pmatrix} 1 \\ 2 \\ -4 \end{pmatrix} \quad V_N := -\int_{P_0}^{N_0} 6x^2\,dx - \int_{P_1}^{N_1} 6y\,dy - \int_{P_2}^{N_2} 4\,dz + 2 \quad V_N = 19$$

Potential Field of a System of Charges: Conservative Prop.

Problem:

Find the absolute potential in the presence of field due to system of point charges Q_i.



$$V(\boldsymbol{r}) = \frac{Q}{4\pi\varepsilon_0 r}$$

Absolute Potential due to point charge Q at origin.

Let Q_1 is located at $\boldsymbol{r}_1$ (source point) instead of origin, then potential at $\boldsymbol{r}$ (field point) due to $Q_1(\boldsymbol{r}_1)$ is $V(\boldsymbol{r})$ and involves the distance between $\boldsymbol{r}$ and $\boldsymbol{r}_1$,

$$V(\boldsymbol{r}) = \frac{Q_1}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_1|}$$

The potential due to another charge $Q_2(\boldsymbol{r}_2)$ becomes,

$$V(\boldsymbol{r}) = \frac{Q_1}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_1|} + \frac{Q_2}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_2|}$$

Continue adding charges, the potential arising from n point charges is,

$$V(\boldsymbol{r}) = \sum_{i=1}^{n} \frac{Q_i}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_i|}$$

Potential Field of a System of Charges: Conservative Prop.

Problem:

Absolute potential in the presence of field due to system of point charges (both Discrete and Continuous).

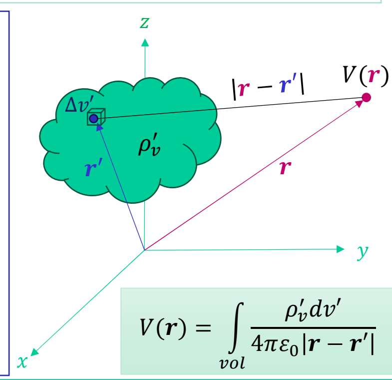
Relationship for Potential due to Discrete Point Charges,

$$V(\boldsymbol{r}) = \sum_{i=1}^{n} \frac{Q_i}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_i|}$$

Assuming cloud of charges having density ρ_v' [C/m^3], a point charge in a cloud $\Delta Q_i = \rho_v'\Delta v_i$ relates to potential as,

$$V(\boldsymbol{r}) = \lim_{i\to\infty} \sum_{i} \frac{\rho_v'\Delta v_i}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}_i|}$$

Relationship for Potential due to Continuous Charge Distribution,



$$V(\boldsymbol{r}) = \int_{vol} \frac{\rho_v' dv'}{4\pi\varepsilon_0|\boldsymbol{r}-\boldsymbol{r}'|}$$

The potential $V(\boldsymbol{r})$ is work done in carrying a unit positive charge from infinite to the field point $\boldsymbol{r}$ depends on the total charge present in the cloud, i.e., $\int_{vol} \rho_v(\boldsymbol{r}')dv'$ and the distance from the source point to the field point, $|\boldsymbol{r}-\boldsymbol{r}'|$.

Summary:

Absolute Potential due to System of Charges

Charge	Potential	Field
Point Charge, Q	$V(\boldsymbol{r}) = \dfrac{Q}{4\pi\varepsilon_0 r}$	$\mathbf{E} = \dfrac{Q}{4\pi\varepsilon_0 r^2}\boldsymbol{a}_r$
Discrete Point Charges, Q_i	$V(\boldsymbol{r}) = \sum_{i=1}^{n} \dfrac{Q_i}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}_i\rvert}$	
Continuous Point Charges, ρ_v	$V(\boldsymbol{r}) = \int_{vol} \dfrac{\rho_v' dv'}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}'\rvert}$	$\mathbf{E}(\boldsymbol{r}) = \int_{vol} \dfrac{\rho_v' dv'}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}'\rvert^2} \dfrac{\boldsymbol{r}-\boldsymbol{r}'}{\lvert\boldsymbol{r}-\boldsymbol{r}'\rvert}$
Line Charge, ρ_L	$V(\boldsymbol{r}) = \int \dfrac{\rho_L' dL'}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}'\rvert}$	
Surface Charge, ρ_S	$V(\boldsymbol{r}) = \int_S \dfrac{\rho_S' dS'}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}'\rvert}$	

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Example 4.3: Find V on the z-axis for a uniform line charge ρ_L in the form of a ring of radius $\rho = a$ and located at $z = 0$ plane.

$$V(\boldsymbol{r}) = \int \frac{\rho_L' dL'}{4\pi\varepsilon_0 \lvert\boldsymbol{r}-\boldsymbol{r}'\rvert}$$

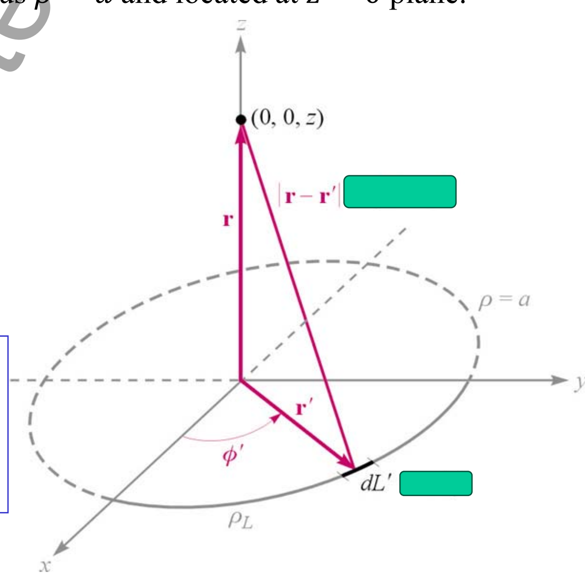
$\boldsymbol{r} = z\boldsymbol{a}_z$ $\quad$ $\boldsymbol{r}' = a\boldsymbol{a}_\rho$

$\boldsymbol{r}-\boldsymbol{r}' = z\boldsymbol{a}_z - a\boldsymbol{a}_\rho$

$\lvert\boldsymbol{r}-\boldsymbol{r}'\rvert = \sqrt{a^2+z^2}$

$$V(\boldsymbol{r}) = \int_{\varphi=0}^{2\pi} \frac{\rho_L a d\varphi'}{4\pi\varepsilon_0\sqrt{a^2+z^2}}$$

$$V(\boldsymbol{r}) = \frac{\rho_L a}{2\varepsilon_0\sqrt{a^2+z^2}} \quad \text{[V]}$$



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